PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Md Ektearul Haque Seajon
Program/Batch	B084-Cybersecurity Program
Date	10 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM3 (Maltego Scanning) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I performed the activities documented in this report only within the authorized scope of the cybersecurity practical assessment and on my own local network. The activities were conducted for educational and research purposes.

The information and findings presented in this report are based on the actual outputs collected during the exercises. No unauthorized exploitation, modification, or disruption of systems was performed.

All testing activities must be conducted only against systems and networks for which appropriate authorization has been obtained. The author, instructors, and training organization are not responsible for any unauthorized or unlawful use of the techniques described in this report.

2. Introduction

This report documents the practical activities performed during the Footprinting, Reconnaissance, and Network Scanning phases of the cybersecurity training program.

The first activity involved performing reconnaissance against the **networkwalks.com** domain using multiple Kali Linux tools. The objective was to collect publicly available information about the target domain, identify its web technologies, inspect DNS records, examine HTTP response headers, and determine whether a Web Application Firewall (WAF) was present.

The second activity involved performing network discovery on my own local LAN using Zenmap (Nmap GUI). The objective was to identify the local network subnet, discover active hosts, collect available IP and MAC address information, and visualize the discovered network devices through the Zenmap topology feature.

The following tools were used during the practical exercises:

- WHOIS
- WhatWeb
- Nslookup
- Curl
- Wafw00f
- DNSRecon
- Maltego
- Zenmap (Nmap GUI)

All findings documented in this report are derived from the actual command-line outputs and screenshots collected during the exercises.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tools	Purpose
Kali Linux	Operating system used for reconnaissance activities
WHOIS	Collect domain registration, registrar, and name server information
WhatWeb	Identify website technologies, CMS, server information, and plugins
Nslookup	Resolve domain names into IP addresses using DNS
Curl	Inspect HTTP response headers and server responses
Wafw00f	Identify the presence and type of Web Application Firewall
DNSRecon	Enumerate DNS records, including A, NS, MX, TXT, SOA, and SRV records
Maltego	Visualize relationships between domain-related entities and infrastructure
Zenmap	Discover live hosts and visualize local network topology
Nmap	Perform network discovery through command-line scanning
Windows / Local Network	Provide the environment for internal network discovery

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the target domain **networkwalks.com** using multiple Kali Linux tools. Each tool was used to collect a different category of information about the target's publicly accessible infrastructure.

4.1.1 WHOIS Information Gathering

Command Executed:

```
whois networkwalks.com
```

I used the WHOIS utility to collect publicly available domain registration information about networkwalks.com.

Observations:

- Domain Name: NETWORKWALKS.COM
- Registrar: GoDaddy.com, LLC
- Registrar WHOIS Server: whois.godaddy.com
- Creation Date: 06 November 2019
- Updated Date: 12 November 2025
- Registry Expiry Date: 06 November 2027
- Domain Status: clientTransferProhibited
- Domain Status: clientUpdateProhibited
- Domain Status: clientRenewProhibited
- Domain Status: clientDeleteProhibited
- Name Server 1: NS6135.HOSTGATOR.COM
- Name Server 2: NS6136.HOSTGATOR.COM
- Registrant Organization: Domains By Proxy, LLC

The WHOIS output also showed that the registrant information was protected through a domain privacy service.

Security Significance:

WHOIS information helps security professionals understand domain ownership-related metadata, registrar details, registration history, and DNS infrastructure. This information can support further reconnaissance and infrastructure mapping.

4.1.2 Website Technology Fingerprinting Using WhatWeb

Command Executed:

```
whatweb networkwalks.com
```

I used WhatWeb to identify the technologies and components used by the target website.

Observations:

- Web Server: Apache
- CMS: WordPress 7.1.4
- WordPress Plugin: WP Download Manager 3.3.76
- JavaScript Library: jQuery 3.7.1
- Framework: Bootstrap 7.1.4 (as identified by WhatWeb)
- Website IP Address: 192.232.216.135
- Country: United States
- Email Address: info@networkwalks.com

- Google Tag Manager: Detected
- Website Title: Networkwalks Academy
- HTTP Response: 200 OK

WhatWeb also identified several HTTP response headers, including permissions-policy, link, upgrade, referrer-policy, and x-nginx-cache.

Security Significance:

Technology fingerprinting helps identify the software components and technologies used by a website. Attackers may use publicly exposed version information to research potential weaknesses in outdated software or plugins.

The identified technologies should be reviewed against official security advisories to determine whether any known vulnerabilities are applicable. The version information alone does not confirm a vulnerability.

4.1.3 DNS Resolution Using Nslookup

Command Executed:

```
nslookup networkwalks.com
```

I used Nslookup to resolve the target domain name into its corresponding IP address.

Observations:

- DNS Server: 8.8.8.8
- DNS Server Address: 8.8.8.8#53
- Domain: networkwalks.com
- Resolved IP Address: 192.232.216.135
- Response Type: Non-authoritative Answer

Security Significance:

DNS resolution provides the IP address associated with a domain name. This information is useful for identifying the network endpoint of a website and supporting further authorized network reconnaissance.

4.1.4 HTTP Header Analysis Using Curl

Command Executed:

```
curl -I https://networkwalks.com
```

I used Curl with the -I option to retrieve the HTTP response headers from the target website.

Observations:

- HTTP Response: HTTP/2 200
- Server: Apache
- Content-Type: text/html; charset=UTF-8

- X-Nginx-Cache: WordPress
- X-Endurance-Cache-Level: 0
- Referrer-Policy: no-referrer-when-downgrade
- Permissions-Policy: Present
- Link Header: WordPress REST API endpoint identified
- REST API Endpoint: /wp-json/

The response also included a WordPress-related Set-Cookie header with HttpOnly and Secure attributes.

Security Significance:

HTTP response header analysis helps identify server information, application behavior, security-related headers, and publicly exposed application endpoints.

The presence of the WordPress REST API endpoint can assist further application enumeration. However, its presence is not itself a security vulnerability.

4.1.5 Web Application Firewall Detection Using Wafw00f

Command Executed:

```
wafw00f networkwalks.com
```

I used Wafw00f to identify whether a Web Application Firewall was protecting the target website.

Observations:

- Target Website: https://networkwalks.com
- Detection Status: Web Application Firewall Detected
- Identified WAF: ModSecurity (SpiderLabs)
- Number of Requests: 2
- Tool Version: Wafw00f v2.4.2

The tool identified that the target website was behind a ModSecurity (SpiderLabs) Web Application Firewall.

Security Significance:

A Web Application Firewall helps monitor and filter HTTP requests to protect web applications from malicious traffic and common web-based attacks.

Identifying the WAF provides information about the target's existing security architecture. Further authorized testing would be necessary to assess its effectiveness.

4.1.6 DNS Enumeration Using DNSRecon

Command Executed:

```
dnsrecon -d networkwalks.com
```

I used DNSRecon to enumerate DNS records associated with the target domain.

Observations:

The DNSRecon output identified the following records:

SOA Record:

- Primary Name Server: ns6135.hostgator.com
- SOA IP Address: 50.87.144.87

NS Records:

- ns6135.hostgator.com – 50.87.144.87
- ns6136.hostgator.com – 192.232.216.131

MX Record:

- mail.networkwalks.com – 50.87.144.87

A Record:

- networkwalks.com – 192.232.216.135

TXT Records:

- SPF Record: v=spf1 a +mx +ip4:50.87.144.87 include.com ~all
- Google Site Verification TXT record detected.

SRV Records:

DNSRecon also identified multiple SRV records associated with the _autodiscover._tcp subdomain. These records pointed to cPanel email discovery services.

Enumeration Summary:

- Total Records Found: 8
- Enumeration Status: Completed

Security Significance:

DNS enumeration provides useful information about domain infrastructure, mail services, name servers, and related services.

The collected DNS records can help security professionals understand the organization's publicly visible infrastructure and identify services that may require additional security review.

4.1.7 Domain Relationship Mapping Using Maltego

I used Maltego Desktop to visualize relationships associated with the networkwalks.com domain.

Observations:

The Maltego graph displayed several related entities, including:

- networkwalks.com
- NETWORKWALKS.COM
- GoDaddy.com, LLC
- VeriSign
- VeriSign Global Registry Services
- whois.godaddy.com
- WHOIS
- RDAP Service
- abuse@godaddy.com
- info@networkwalks.com
- Registrar-related contact information
- Related phone number: 480-624-2505

The graph represented relationships between the target domain, registrar, registry services, WHOIS services, and contact-related entities.

Security Significance:

Maltego helps security professionals visually understand relationships between publicly available information and related infrastructure entities.

This type of relationship mapping supports OSINT investigations and provides a broader understanding of the target's publicly exposed domain information.

4.2 Network Scanning with Zenmap

For the second practical activity, I performed network discovery on my own local LAN using Zenmap, the graphical interface of Nmap.

The main objective was to identify active devices within the local network and visualize their network relationships.

4.2.1 Local Network Discovery

Target Subnet:
`192.168.1.0/24`

Scan Profile: Ping Scan

Nmap Command:
`nmap -sn 192.168.1.0/24`

I selected the Ping Scan profile in Zenmap to identify active hosts within the specified subnet.

Scan Results:

- Total IP Addresses Scanned: 256
- Live Hosts Discovered: 3
- Scan Duration: 3.07 seconds
- Nmap Version: 7.99

Discovered Hosts:

Host IP Address	MAC Address	Vendor / Identification
192.168.1.1	52:55:C0:A8:01:01	Unknown
192.168.1.3	08:00:27:E1:CE	Oracle VirtualBox Virtual NIC
192.168.1.6	Not displayed in captured output	Not identified

The scan successfully identified three active hosts in the local network.

4.2.2 Network Topology Visualization

After completing the Ping Scan, I opened the Topology section in Zenmap to visualize the discovered devices.

Observations:

- Three live hosts were displayed in the topology.
- The central node represented localhost.
- The discovered devices were displayed around the central node.
- The three discovered IP addresses were:
    - 192.168.1.1
    - 192.168.1.3
    - 192.168.1.6

The topology view provided a graphical representation of the discovered local network devices and their relationship with the scanning machine.

Security Significance:

Network topology visualization helps administrators understand the devices visible within a local network. It can support network inventory management, device identification, and the investigation of unexpected hosts.

Note: The Ping Scan was used for host discovery only. It did not perform a full TCP/UDP port scan or establish whether any specific service was vulnerable.

5. Risk Analysis / Impact

Based on the reconnaissance and network scanning activities, I identified the following security observations.

These findings represent information exposure and network visibility observations. They are not confirmed exploitable vulnerabilities.

Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
Publicly available domain registration information	WHOIS identified registrar, name servers, and domain metadata	May assist attackers in infrastructure reconnaissance	Low
Website technology information exposed	WhatWeb identified WordPress, Apache, jQuery, and plugin information	May help attackers identify software versions requiring security review	Medium
Publicly identifiable server IP	Nslookup resolved the domain to 192.232.216.135	May support further infrastructure enumeration	Low
HTTP technical information exposed	Curl identified server headers and WordPress REST API endpoint	May assist application fingerprinting and further enumeration	Low
WAF technology identifiable	Wafw00f detected ModSecurity (SpiderLabs)	Reveals information about the website's security architecture	Low
DNS infrastructure information exposed	DNSRecon identified A, NS, MX, TXT, SOA, and SRV records	May help attackers map domain and mail infrastructure	Medium
Domain relationship information exposed	Maltego displayed registrar, registry, WHOIS, and contact relationships	May assist OSINT-based reconnaissance	Low
Multiple active devices visible on local network	Zenmap identified three live hosts in 192.168.1.0/24	Unknown or unauthorized devices may potentially remain undetected without inventory verification	Medium

Risk Level Key:

- Critical – Severe security impact
- High – Significant security impact
- Medium – Moderate security concern
- Low – Limited security concern

The risk levels above are preliminary assessments of the observations collected during the practical exercises.

The presence of publicly available technical information does not automatically indicate a vulnerability. No exploitation or vulnerability validation was performed during these activities.

6. Recommendations

Based on the observations collected during the exercises, the following security improvements are recommended:

1. Review Publicly Exposed Domain Information

Organizations should periodically review publicly accessible domain registration information and ensure that unnecessary sensitive contact details are not exposed.

2. Keep Website Technologies Updated

WordPress, plugins, JavaScript libraries, and other website components should be updated regularly. Software versions should be checked against official security advisories.

3. Review Server Information Exposure

Organizations should review publicly exposed server information and remove unnecessary technical details from HTTP responses where appropriate.

4. Secure HTTP Response Headers

Security-related HTTP headers should be reviewed and configured according to the application's requirements and security best practices.

5. Maintain Proper WAF Configuration

The detected ModSecurity WAF should remain properly configured, updated, and monitored. Its rules should be reviewed regularly to ensure effective protection against web-based threats.

6. Review DNS Records Regularly

DNS records should be reviewed periodically to ensure that only required services and records are publicly exposed.

7. Monitor Email and DNS Infrastructure

Mail-related DNS records, SPF configuration, and service discovery records should be reviewed to maintain proper email security and infrastructure configuration.

8. Perform Regular Internal Network Discovery

Organizations should periodically perform authorized network discovery to identify active devices within their internal networks.

9. Investigate Unknown Devices

All discovered devices should be compared against the organization's authorized asset inventory. Unknown devices should be investigated before being classified as unauthorized.

10. Maintain Network Documentation

Network IP addresses, MAC addresses, device identities, and network topology should be documented and updated regularly.

11. Perform Further Authorized Security Testing

Additional vulnerability assessment and penetration testing should be conducted within an approved scope to validate any potential security weaknesses identified during reconnaissance.

7. Conclusion

During the cybersecurity practical exercises, I completed activities covering footprinting, reconnaissance, OSINT-based relationship mapping, and local network discovery.

In the reconnaissance phase, I used WHOIS to collect domain registration information, WhatWeb to identify website technologies, Nslookup to resolve the target domain, Curl to inspect HTTP response headers, Wafw00f to identify the Web Application Firewall, and DNSRecon to enumerate DNS records.

I also used Maltego to visualize relationships between the target domain and its associated registrar, registry, WHOIS services, and other publicly available entities.

In the network scanning phase, I used Zenmap to perform a Ping Scan against my own local network using the subnet 192.168.1.0/24. The scan identified three live hosts from 256 scanned IP addresses. I also collected the available MAC address information and generated a network topology visualization.

These practical exercises helped me understand how reconnaissance and network discovery are performed during the initial stages of a penetration testing engagement.

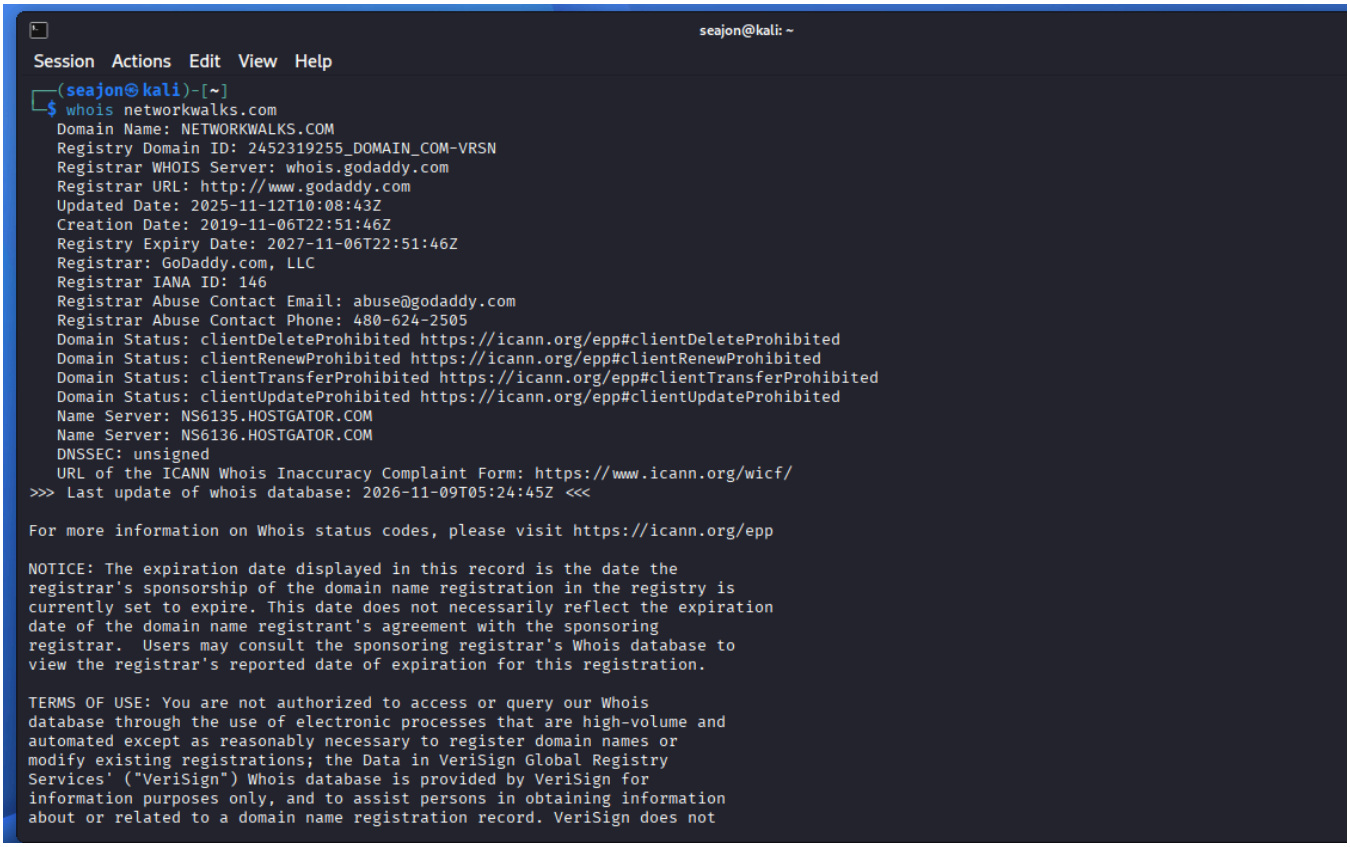
I learned that publicly available information can provide valuable insights into an organization's infrastructure before conducting more advanced security assessments.

Finally, this exercise improved my practical understanding of Kali Linux reconnaissance tools, OSINT techniques, DNS enumeration, HTTP header analysis, and network discovery using Nmap and Zenmap.

8. Evidences Collected

The following screenshots were collected as evidence of the practical activities performed during the assessment.

WHOIS:



```
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T05:08:41Z
Creation Date: 2019-11-06T17:51:46Z
Registrar Registration Expiration Date: 2027-11-06T17:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
Registrant Street: 100 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
```

WhatWeb:

```
┌──(seajon㉿kali)-[~]
└─$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]

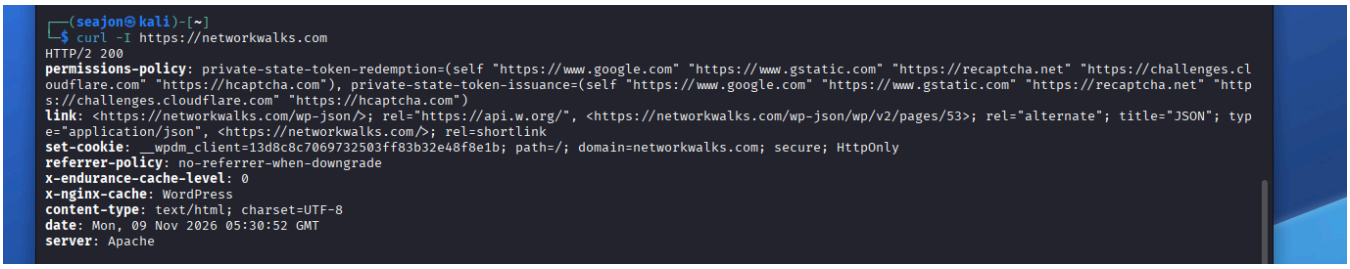
┌──(seajon㉿kali)-[~]
└─$
```

Nslookup:

```
┌──(seajon㉿kali)-[~]
└─$ nslookup networkwalks.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

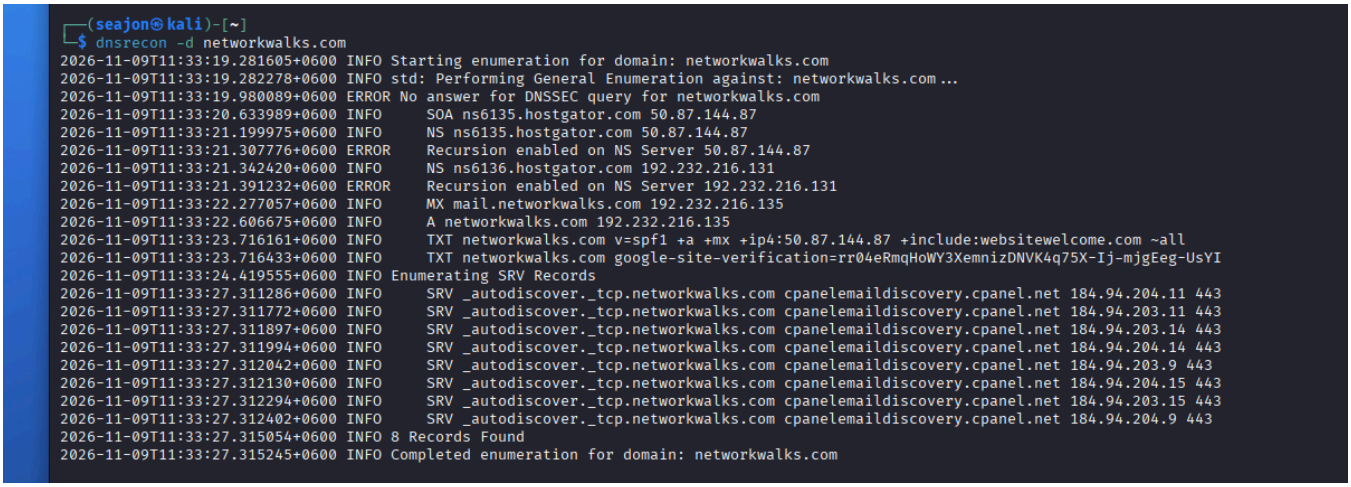
Curl:



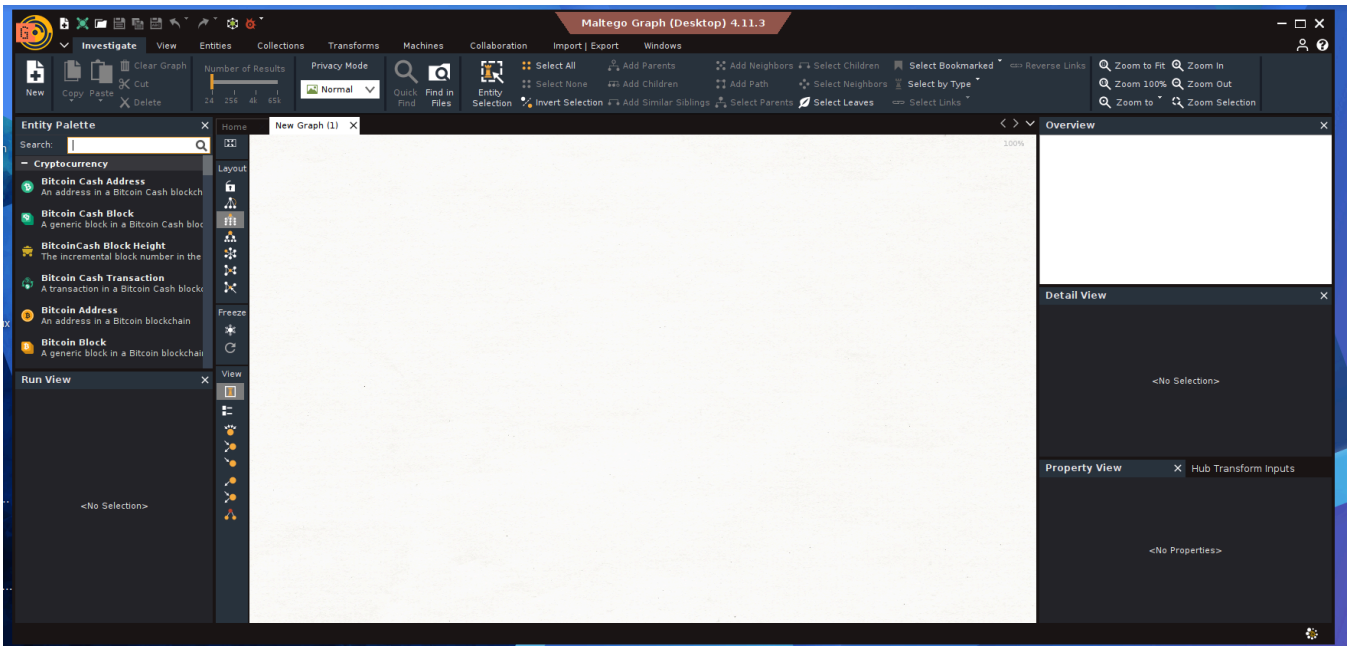
Wafw00f:

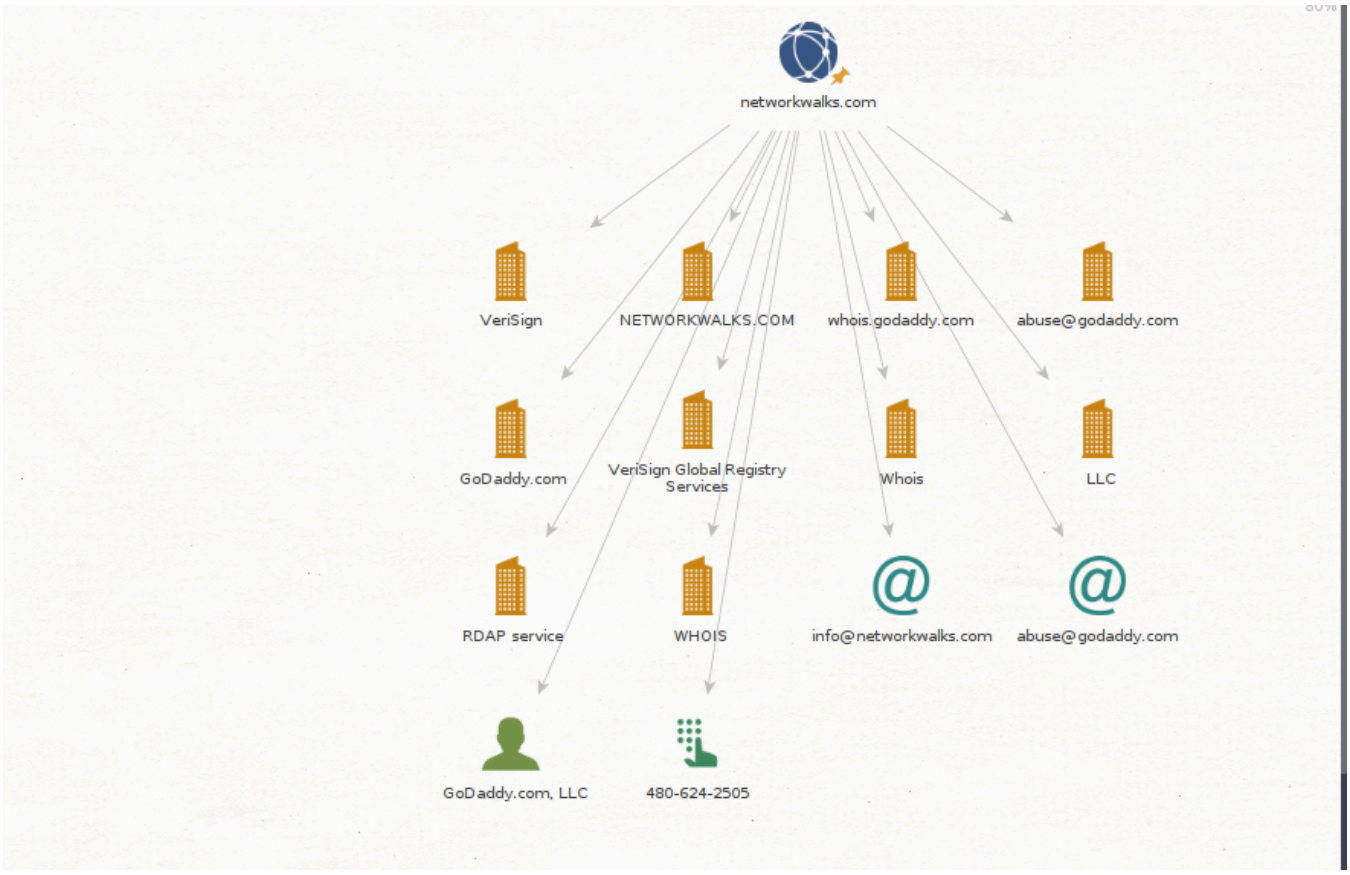


DNSRecon:

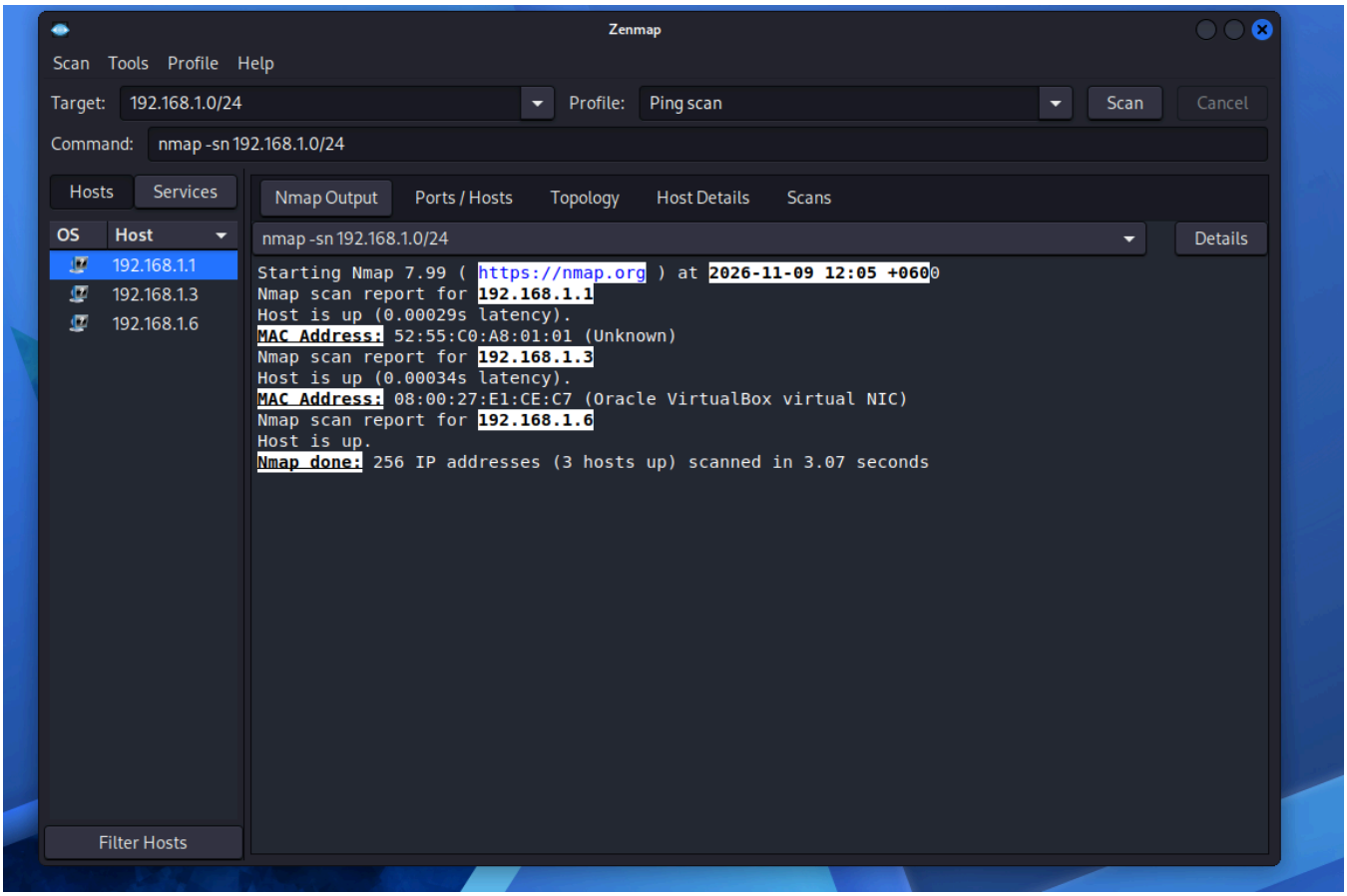


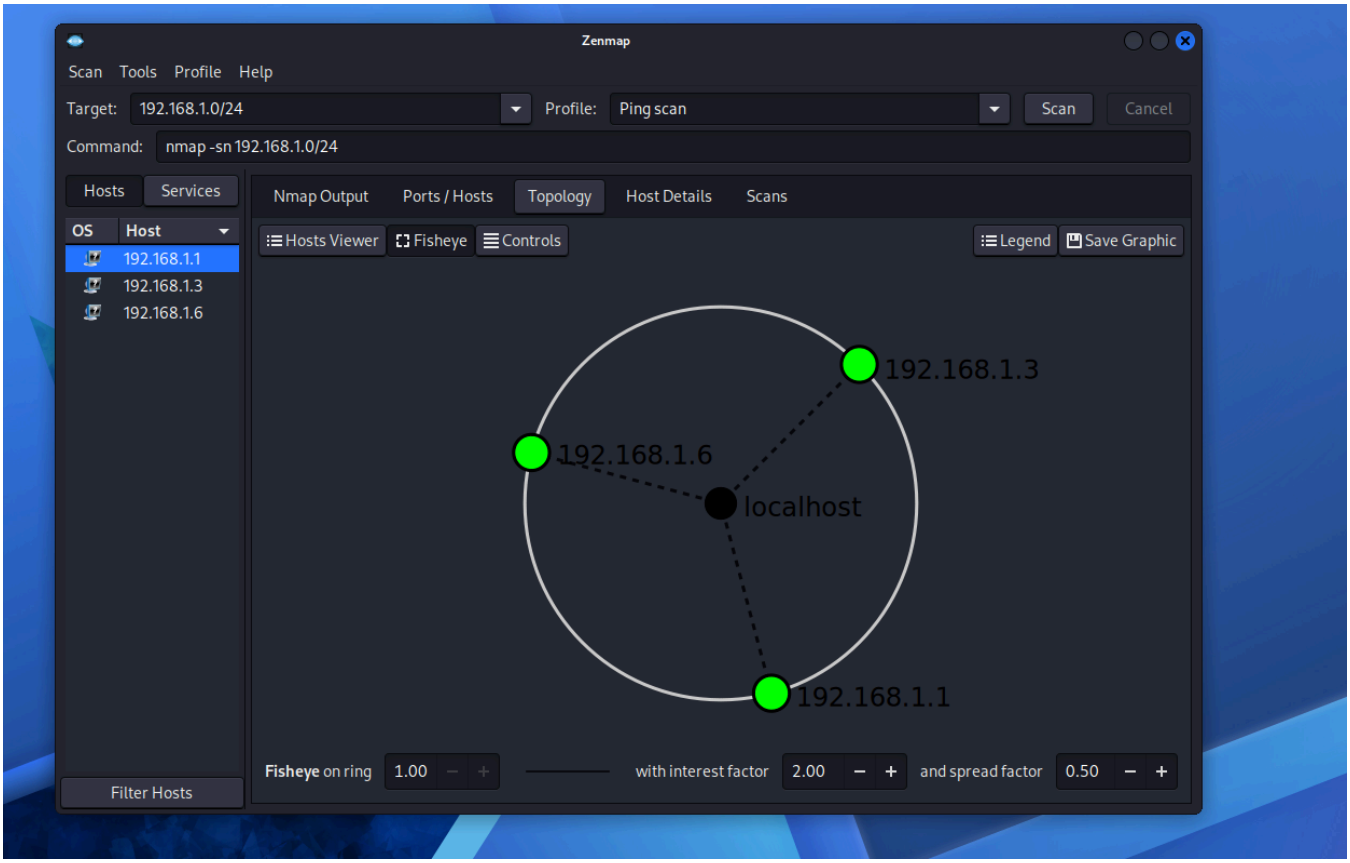
Maltego:


Maltego Graph (Desktop) 4.11.3
Investigate
View
Entities
Collections
Transforms
Machines
Collaboration
Import | Export
Windows
Entity Palette
Cryptocurrency
Bitcoin Cash Address
Bitcoin Cash Block
BitcoinCash Block Height
Bitcoin Cash Transaction
Bitcoin Address
Bitcoin Block
Run View
<No Selection>
New Graph (1)
Overview
Detail View
<No Selection>
Property View
Hub Transform Inputs
<No Properties>


networkwalks.com
VeriSign
NETWORKWALKS.COM
whois.godaddy.com
abuse@godaddy.com
GoDaddy.com
VeriSign Global Registry Services
Whois
LLC
RDAP service
WHOIS
info@networkwalks.com
abuse@godaddy.com
GoDaddy.com, LLC
480-624-2505

Zenmap:


Zenmap
Scan Tools Profile Help
Target: 192.168.1.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 192.168.1.0/24
Hosts
Services
OS
Host
192.168.1.1
192.168.1.3
192.168.1.6
Filter Hosts
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn 192.168.1.0/24
Details
Starting Nmap 7.99 (https://nmap.org) at 2026-11-09 12:05 +0600
Nmap scan report for 192.168.1.1
Host is up (0.00029s latency).
MAC Address: 52:55:C0:A8:01:01 (Unknown)
Nmap scan report for 192.168.1.3
Host is up (0.00034s latency).
MAC Address: 08:00:27:E1:CE:C7 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.6
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.07 seconds



---End of Report---

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📌 **Project Information**
Project Name: Footprinting & Network Scanning | **Week:** 02